

Department of Mathematics and Applied Mathematics

3MS Metric Spaces

About suprema and infima in the reals

This is a brief summary of how one works with suprema and infima of real numbers, with some comments on misconceptions. This all comes under the heading of “Real Analysis,” so should be familiar from last year.

Definitions Let A be any non-empty subset of $\mathbb{R}$, and $u, \ell, x \in \mathbb{R}$.

We say that u is an **upper bound** of A iff $u \geq a$ for all $a \in A$.

So then x is **not an upper bound** of A iff there is some $a_0 \in A$ such that $x < a_0$.

Similarly, ℓ is a **lower bound** of A iff $\ell \leq a$ for all $a \in A$.

So then x is **not a lower bound** of A iff there is some $a_0 \in A$ such that $x > a_0$.

A subset of $\mathbb{R}$ that has an upper bound is called bounded above.

A subset of $\mathbb{R}$ that has a lower bound is called bounded below.

A subset of $\mathbb{R}$ that has an upper bound and a lower bound is called bounded.

Let A be a non-empty subset of $\mathbb{R}$ that is bounded above. The smallest upper bound of A is called the **supremum** of A , and is denoted by $\sup A$.

Let A be a non-empty subset of $\mathbb{R}$ that is bounded below. The greatest lower bound of A is called the **infimum** of A , and is denoted by $\inf A$.

(It is an axiom of the real number system that every non-empty bounded set has a supremum and infimum.)

Maximum vs. supremum

If A is a subset of $\mathbb{R}$, A may or may not have a largest element. If it does, this element is called the **maximum** of A ; written $\max A$. Similarly, if it has a smallest element, that is denoted $\min A$, for the **minimum** of A . It would be very convenient if every set which is bounded above has a maximum, but this is simply not true. (Take the open interval $(0, 6)$, for instance.) This is why we need the notion of supremum. We list a few useful things to remember; in the following, A is a subset of $\mathbb{R}$:

1. If the set A has a maximum, $\max A = \sup A$.
If the set A has a minimum, $\min A = \inf A$.
2. If A has only finitely many elements, then A has a maximum and a minimum (and, obviously $\max A = \sup A$ and $\min A = \inf A$).
3. There exist sets A for which $\sup A$ exists, but $\max A$ does not. (The open interval $(0, 6)$ is such a set.)
4. There exist sets A for which $\sup A$ does not exist. ($\mathbb{N}$ is such a set.)

How to show $\sup A = s$:

Lemma Let A be a non-empty subset of $\mathbb{R}$ and $s \in \mathbb{R}$. Then $s = \sup A$ iff

1. $s \geq a$ for all $a \in A$ (i.e. s is an upper bound of A) **and**
2. if $u \geq a$ for all $a \in A$ then $u \geq s$ (i.e. if u is an upper bound of A , then u is greater than or equal to s).

Lemma Let A be a non-empty subset of $\mathbb{R}$ and $s \in \mathbb{R}$. Then $s = \sup A$ iff

1. $s \geq a$ for all $a \in A$ (i.e. s is an upper bound of A) **and**
2. if $x < s$ then there exists $a_0 \in A$ with $x < a_0$ (i.e. if x is less than s then x is not an upper bound of A .)

How to show $\sup A \leq s$:

Show $a \leq s$ for all $a \in A$.

Obviously, corresponding results can be written out for infima.

Examples in metric spaces

1. Clearly, we can show $\inf C \geq b$ by showing that $c \geq b$ for all $c \in C$. As an example of this, look at Tutorial 3, Question 4(a). You were asked to show that if $C \subseteq D$ then $\inf C \geq \inf D$:
Since $\inf D \leq d$ for all $d \in D$, we also have $\inf D \leq c$ for all $c \in C$. So $\inf D \leq \inf C$. That's the entire argument.
2. In the above, we explained the difference between maximum and supremum. A similar difference exists between minimum and infimum. Here is a misconception, based on not understanding that difference: Look at Tutorial 2, Question 2 or Tutorial 3, Question 4. In the proofs, you cannot assume that there exist $a \in A, b \in B$ such that $\text{dist}(A, B) = d(a, b)$. That would be assuming that the infimum is actually a minimum.

3. Here's a new example, to see if you are comfortable with all this.

Suppose (X, d) is a metric space and Y is a subset of X . If the supremum $D = \sup\{d(y_1, y_2) \mid y_1, y_2 \in Y\}$ exists, we say that the set Y has **finite diameter**, in fact we say it has **diameter** D . If the supremum does not exist, we say Y has **infinite diameter**.

Lemma In any metric space (X, d) , if A and B are subsets of X with finite diameter, and $A \cap B \neq \emptyset$, then

$$\text{diam}(A \cup B) \leq \text{diam } A + \text{diam } B.$$

Proof

If $r, s \in A$, then $d(r, s) \leq \sup \{d(a, a') : a, a' \in A\} = \text{diam } A \leq \text{diam } A + \text{diam } B$. Similarly, if $r, s \in B$, then $d(r, s) \leq \sup \{d(b, b') : b, b' \in B\} = \text{diam } B \leq \text{diam } A + \text{diam } B$.

Take $t \in A \cap B$. If $r \in A$ and $s \in B$, then $d(r, s) \leq d(r, t) + d(t, s) \leq \sup \{d(a, a') : a, a' \in A\} + \sup \{d(b, b') : b, b' \in B\} = \text{diam } A + \text{diam } B$.

A similar argument holds if $r \in B$ and $s \in A$.

In total, we have shown that for all $r, s \in A \cup B$, $d(r, s) \leq \text{diam } A + \text{diam } B$.

So $\text{diam}(A \cup B) \leq \text{diam } A + \text{diam } B$.